Hassan Saei, MSc. PhD.

Imagine Institute of Genetic Diseases
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Email — Personal website — GitHub — LinkedIn

PROFESSIONAL PROFILE

Postdoctoral researcher in RNA and gene therapy with strong experience in both wet-lab and computational analysis. Interested in translational research aimed at improving genetic diagnosis and developing and testing novel therapeutic strategies in complex in vitro disease models. Early in my PhD, I developed a bioinformatics tool that significantly improved genetic diagnosis in patients with ADTKD-MUC1 using short-read sequencing data. I subsequently focused on developing physiologically relevant in vitro kidney organoid models to evaluate therapeutic approaches. My research integrates genomics, computational analysis, and organoid models to accelerate translational discovery and precision medicine.

EDUCATION

Université Paris Cité, Institut Pasteur, France *2021-2024*
Ph.D. in Genetics
Bio Sorbonne Paris Cite, DGNRV department
Fellowship from the PPU-Imagine Program
Thesis: Characterization of missing variants in hereditary renal diseases and development of RNA-based therapy in kidney organoids [Link]

Iran University of Medical Sciences, Iran *2017-2020*
M.Sc. in Human Genetics
Overall GPA: 4/4

University of Tabriz, Iran *2013-2017*
B.Sc. in Biology
Overall GPA: 3.84/4

*All credentials are evaluated by World Education Services [Link]

PROFESSIONAL EXPERIENCES

Imagine Institute of Genetic Diseases, Paris, France Jan 2025 - Present
Postdoctoral Researcher *Paris, France*

- Research Focus: Developing engineered AAVs for targeted therapy development for podocytopathies in partnership with Sanofi, and WhiteLab Genomics (WIDGeT consortium project)
- Developing *in vitro* and *in vivo* AAV binding and transduction screening pipelines and their computational analysis

Oct 2021 - Dec 2024
Ph.D. Candidate *Paris, France*

- Research Focus: Improving genetic diagnosis of hereditary renal diseases and developing robust disease models for therapeutic discovery
- Developed a splice-modulation therapeutic approach for Alport syndrome using human kidney organoids
- Developed a computational pipeline for variant detection in the coding-VNTR in ADTKD-MUC1 using short-read sequencing data (vntyper-online)
- Integrated genomic and transcriptomic analyses to enhance the genetic diagnosis of X-linked Alport syndrome
- I visited Prof. Rachel Lennon's lab at the University of Manchester, UK, for three weeks and performed proteomics analysis on kidney organoids.

Iran University of Medical Sciences Oct 2018 - Sep 2020
M.Sc. Internship *Tehran, Iran*

- Research Focus: Rare disease genetic diagnosis with NGS, Functional characterization of complex disease
- Modeling pre-eclampsia using 2D culture, biomarker identification and their functional analysis
- Exome sequencing for diagnosis of patients with MSUD
- Bioinformatic analysis via *in house* pipelines for variant detection, classification and *in silico* analysis of the new variants

ANALYTIC AND COMPUTATIONAL SKILLS

Programming Languages	Python, R/Markdown, Bash, High Performance Computing, Git
Workflow Management	Docker, Singularity
Big Data Analysis	RNA-seq, scRNA-seq, scATAC-seq, Proteomics
Sequence analysis	GATK, deepVariant, CADD, SIFT, MutationTaster, IGV
Stem cell research	Human iPSC culture and differentiation
Disease models	Organoid development, Humanized mouse model

PEER-REVIEWED PUBLICATIONS

1. **Saei H**, Ardin B., Kaiser N., Ouahmane M., Gribouval O., Moriniere V., Mautret-Godefroy M., Wopperer F., Riedhammer K., Gale D., Sadeghi-Alavijeh O., Goursaud C., Grunewald O., Prosper C., Lebreton L., Rabant M., Bergmann C., Kormann R., Decramer S., Knebelmann B., Antignac C., Wiesener M., Dorval G. **Mapping Intronic Mutational Hotspots by in Silico Mutagenesis Enables Single Antisense Oligonucleotide Correction of Multiple Variants**. *medRxiv; Under review in Kidney International*. 2026. [Full text]

2. Ouahmane M., **Saei H**., Caux A., Kachmar J., Ait Arkoub Z., Fercocq M., Mautret-Godefroy M., Gribouval O., Morinière V., Knebelmann B., Heidet L., Antignac C., Dorval G. **The frameshifted VNTR length determines the kidney survival in ADTKD-MUC1**. *medRxiv; Under review in Kidney International*. 2026. [Full text]

3. Caux A., Kachmar J., Bengoa J., **Saei H**., Mautret-Godefroy M., Planté-Bordeneuve P., Morinière V., Heidet L., Dorval G. **Functional Identification of Multiple Aberrant Splicing Events Linked to Intronic PKD1 Variant**. *Kidney Medicine*. 2026. [Full text]

4. Popp B, **Saei H**. **MucOneUp: A Simulation Framework for *MUC1*-VNTR Variant Benchmarking**. *BioRxiv*. 2026. [Full text]

5. Popp B[*], **Saei H**[*], Teltsh O., Janoušek V., Přistoupilová A., Vrbacká A., Hartmannová H., Kidd K., Helmuth J., Bleyer A. J., Wiesener M., Fausch K., Rowan C., El Hassan E., Clince M., Cavalleri G., Locher M., Eckardt K.-U., Richter-Pechanska P., ADTKD-Net Consortium, Kmoch S, Antignac C, Conlon P, Dorval G, Živná M, Halbritter J. **VNtyper 2 enables open-access short-read genotyping of *MUC1* VNTR variants in ADTKD at high speed**. *medRxiv; Under revision in Kidney International*. 2026. ([*]co-first authors) [Web server]

6. **Saei H**, Estebe B, Gaudin N, Esmailpour M, Haure J, Gribouval O, Arrondel C, Moriniere V, Tian P, Lennon R, Antignac C, Mollet G, Dorval G. **Splice modulation of COL4A5 reinstates collagen IV assembly in an organoid model of Alport syndrome**. *JCI Insight*. 2025. [Full text]

7. Petzold, F, Jeanpierre C, Chen X, Morinière V, Benmerah A, Dorval G, **Saei H**, Heidet L, Antignac C, Saunier S. **Exome Sequencing in a Large Cohort with Ciliopathy-Related Kidney Disease**. *CJASN*. 2025. [Full text]

8. Kachmar J[*], **Saei H**[*], Morinière V, Heidet L, Knebelmann B, Gribouval O, Mautret-Godefroy M, Burtey S, Vuiblet V, Alla A, Ibalanky A, Moranne O, Nizon M, Savenkoff B, Nitschké P, Antignac C, Dorval G.

Phenotypic Heterogeneity of ADTKD-MUC1 Diagnosed Using VNtyper, a Novel Genetic Technique *American Journal of Kidney Diseases*. 2025. (*co-first authors) [Full text]

9. **Saei H**, Masson C, Morinière V, Kachmar J, Heidet L, Gribouval O, Antignac C, Dorval G. **Using VNtyper from whole exome sequencing data to detect pathogenic variants in the MUC1 gene**. *JASN*. 2024. [Full text]
10. Boisson M, Arrondel C, Cagnard N, Morinière V, Arkoub ZA, **Saei H**, Heidet L, Kachmar J, Hummel A, Knebelmann B, Bonnet-Dupeyron MN, Isidor B, Izzedine H, Legrand E, Couarch P, Gribouval O, Bole-Feysot C, Parisot M, Nitschké P, Antignac C, Dorval G. **A wave of deep intronic mutations in X-linked Alport syndrome**. *Kidney Int*. 2023 Aug; 104(2):367-377. [Full text]
11. **Saei H**, Morinière V, Heidet L, Gribouval O, Lebbah S, Tores F, Mautret-Godefroy M, Knebelmann B, Burtey S, Vuiblet V, Antignac C, Nitschké P, Dorval G. **VNtyper enables accurate alignment-free genotyping of MUC1 coding VNTR using short-read sequencing data in autosomal dominant tubulointerstitial kidney disease**. *iScience*. 2023 Jun 17; 26(7):107171. [Full text]
12. Azarmi M, **Saei H**, Vugt I, Thijssen S, Land B, Elburg R, Garssen J, Folkerts G, Hogenkamp A, Braber S. **Co-trimoxazole-induced reproductive toxicity and placental-barrier disruption: impact on cell-cell junctions and ERK signaling pathway**. *Frontiers in Cell and Developmental Biology*. 2026 Jun 8. [Full text]
13. Abiri M*, **Saei H***, Eghbali M, Karamzadeh R, Shirzadeh T, Sharifi Z, Zeinali S. **Maple syrup urine disease mutation spectrum in a cohort of 40 consanguineous patients and in silico analysis of novel mutations**. *Metab Brain Dis*. 2019 Aug; 34(4):1145-1156. (*co-first authors) [Full text]
14. **Saei H**, Govahi A, Abiri A, Eghbali M, Abiri M. **Comprehensive transcriptome mining identified the gene expression signature and differentially regulated pathways of the late-onset preeclampsia**. *Pregnancy Hypertens*. 2021 Aug; 25:91-102. [Full text]
15. Eghbali M, Fatemi KS, Salehpour S, Abiri M, **Saei H**, Talebi S, Olyaei NA, Yassaee VR, Modarressi MH. **Whole-exome sequencing uncovers novel causative variants and additional findings in three patients affected by glycogen storage disease type VI and Fanconi-Bickel syndrome**. *Front Genet*. 2021 Jan 11; 11:601566. [Full text]
16. Jabbarpour N*, **Saei H***, Jabbarpoor Bonyadi MH, Bonyadi M. **Identification of novel cis-mutations in the GJA8 gene in a 3-generation Iranian family with autosomal dominant congenital nuclear cataract**. *Ophthalmic Genet*. 2022 Oct; 43(5):609-614. (*co-first authors)

PEER REVIEW ACTIVITIES

Reviewed manuscripts for *Molecular Genetics & Genomic Medicine*, *NAR Genomics and Bioinformatics*, *Nature Reviews Nephrology*, *Scientific Data*, and *Wellcome Open Research*, *Cell Biochemistry and Function*, *Kidney International Reports*.

STUDENT SUPERVISION

Supervised Haythem Mami (M.Sc. student, Imagine Institute of Genetic Diseases, Jan 2026 - Sep 2026); project: *Development of an integrated transcriptomic cell atlas of human kidney organoids*.

MAIN TRAINING AND CERTIFICATES

Pasteur Institute Oct 2021 - Dec 2024
PhD courses *Paris, France*

- Scientific Integrity (EMBO)
- Laboratory animal experimentation – Designer Diploma
- Bioinformatics program for PhD students: R and Statistics track, Bio-image analysis track

· Bioinformatics track: Unix, Metagenomics, Single cell data analysis, Functional analysis, Proteomics data analysis, Utilities for HTS data analysis, Expression quantification and differential analysis, Variant calling, ChIP-seq data analysis

AWARDS, SCHOLARSHIPS AND FUNDINGS

- Pasteur-Paris University International Ph.D. Fellowship, France (PPU-Imagine, 2021)
- Poster Prize winner, NephGen Symposium, Freiburg, Germany, 2023
- Travel award winner, 14th International Podocyte Conference, PA, USA, 2023
- Silver medal and 2nd place, National Biology Olympiad for Undergraduate Students, Ministry of Higher Education, Iran
- Outstanding Student Award in Basic Science, University of Tabriz, Ministry of Higher Education, Iran
- Iran National Elite's Foundation Conference Travel Award, 2018
- 1st class honour in the B.Sc. (ranked 1/31 students), University of Tabriz, Iran

MEMBERSHIPS AND CONFERENCE PROCEEDINGS

- The American College of Medical Genetics and Genomics (ACMG) — Member, 2023–2024
- The American Society of Human Genetics (ASHG) — Member, 2023–2024
- The European Society of Human Genetics (ESHG) — Member
- *AI-guided AAV capsid engineering and multi-modal screening pipeline for targeted podocyte gene therapy.* Poster presentation, ESGCT Congress, Hamburg, Germany, 2026.
- *A multimodal screening pipeline accelerates the discovery of engineered AAV capsids with podocyte tropism.* Poster presentation, ESGCT Congress, Hamburg, Germany, 2026.
- *Recurrent pseudoexon activation in COL4A5 intron 6 defines a therapeutically actionable hotspot.* Poster presentation, ESHG Conference, Gothenburg, Sweden, 2026. (*Travel Award winner)
- *Organoids as scalable models for splice modulation therapy development in Alport Syndrome.* Poster presentation, ESHG Conference, Milan, Italy, 2025.
- *Modeling X-linked Alport syndrome with deep-intronic variations in kidney organoids for antisense oligonucleotide-based therapy development.* Poster presentation, BaCell3D, Basel, Switzerland, 2025.
- *Developing and characterizing X-linked Alport Syndrome kidney organoid model with deep-intronic variation for drug discovery.* Poster presentation, American Society of Matrix Biology, Basement Membrane Workshop, Manchester, UK, 2024. (Travel award winner)
- *Highly sensitive variant detection in VNTR is possible with boosted exome sequencing: the example of MUC1-related nephropathy.* Poster presentation, ESHG Conference, Berlin, Germany, 2024.
- *Clinically boosted exome-sequencing is necessary for accurate and highly sensitive genetic diagnosis of ADTKD-MUC1.* Oral presentation, European Renal Association, Stockholm, Sweden, 2024.
- *X-linked Alport syndrome: From transcriptomic diagnosis to preclinical assessment of splice-switching oligonucleotide therapy using patient-derived cells and kidney organoids.* ACMG Annual Clinical Genetics Meeting, 2024.
- *MUC1 gene coding-VNTR alignment-free genotyping approach augmented ADTKD diagnosis in a cohort of 3735 patients with hereditary kidney diseases.* ACMG Annual Clinical Genetics Meeting, 2024.

- *Genotyping coding-VNTR in the MUC1 gene using alignment-free method allows genetic diagnosis of MUC1-related autosomal dominant tubulointerstitial kidney disease.* Poster presentation, NephGen Symposium, Freiburg, Germany, 2023. (*Poster Prize winner)
- *Deep-intronic variants in the X-linked Alport Syndrome: from detection to therapeutic hopes.* Oral presentation, Podocyte Meeting, Philadelphia, PA, USA, 2023. (*Travel Award winner, presented online)
- *Homozygosity mapping in maple syrup urine disease patients from Iran: In silico analysis of novel mutations.* Poster presentation, ESHG Conference, Milan, Italy, 2017.

REFERENCES

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- **Guillaume Dorval, M.D., Ph.D.**
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